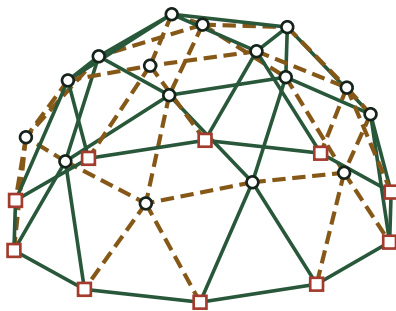


PREPARED FOR JANTZ / BLACK BELT BUILDING

# 12 FT 2V WOOD DOME

Audited geometry + field planning reference

G-000 CANONICAL FRAME AXONOMETRIC - NOT TO SCALE



MEMBER LEGEND      - - - SHORT      — LONG

| REVIEW LAYER        | STATUS             | BOUNDARY                                       |
|---------------------|--------------------|------------------------------------------------|
| Canonical shell     | VERIFIED           | 26 nodes / 65 member axes / 40 faces           |
| Gross wood faces    | VERIFIED           | 30 P1 + 10 P2 node-center templates            |
| Joinery             | SPATIAL CHECK ONLY | Digital clearance; strength and retention open |
| Entrance + platform | CONCEPT ONLY       | Replacement load path and foundation open      |

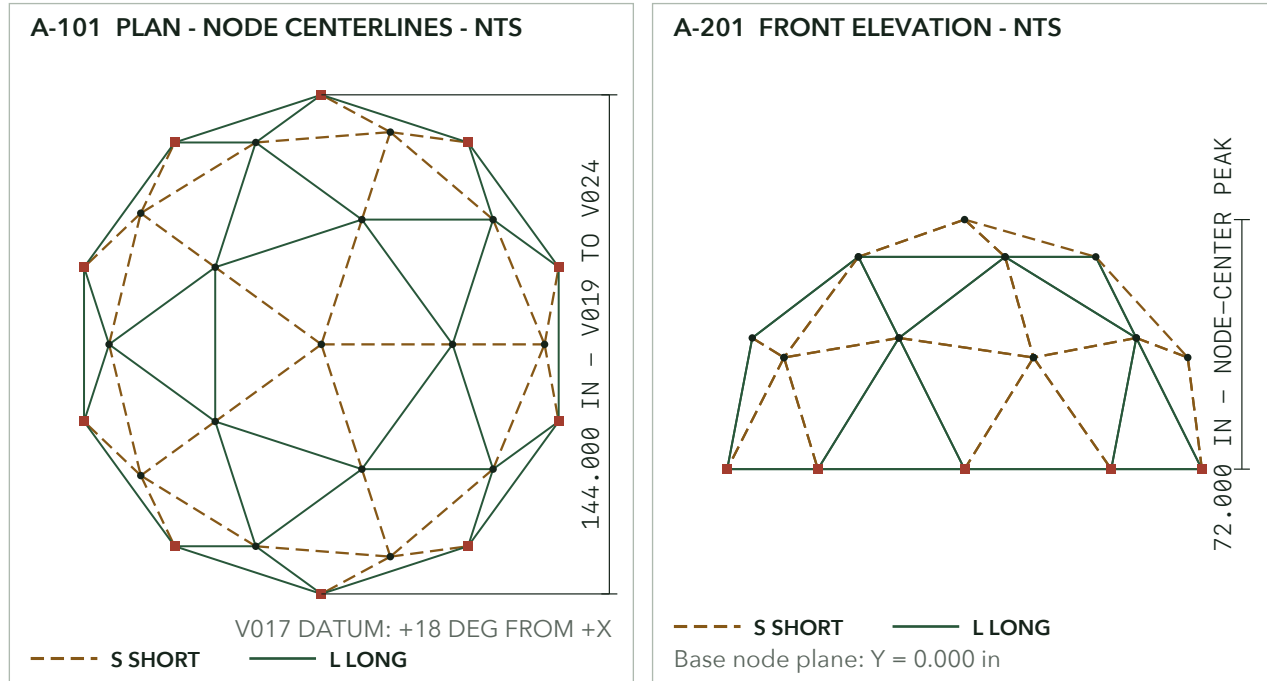
REFERENCE STUDY - NOT FOR FABRICATION, STRUCTURAL, PERMIT, OR OCCUPANCY USE

Print-optimized and text-searchable. Accessible interactive companion: [monroe-domes.vercel.app](https://monroe-domes.vercel.app). This PDF is not represented as PDF/UA tagged.

## G-101 / CANONICAL MODEL

## Plan and elevation

All linework terminates at mathematical node centers. Dimensions describe the computational axes, not timber outside faces or interior clearances.



| PARAMETER     | AUDITED VALUE   | MEANING                                |
|---------------|-----------------|----------------------------------------|
| Sphere radius | 72.000 in       | Center to mathematical node sphere     |
| Base diameter | 144.000 in      | V019 to V024; opposite base nodes      |
| Peak height   | 72.000 in       | Base node plane to apex node           |
| Base          | Regular decagon | 10 coplanar nodes / 10 boundary chords |

Checks:  $V - E + F = 26 - 65 + 40 = 1$ . Port incidence:  $2E = 130 = 10(4) + 6(5) + 10(6)$ . +Y is vertical; the base lies at  $Y = 0$ . V017 is +18 DEG from +X at (68.476069, 0.000000, 22.249224) in.

FIGURE TEXT EQUIVALENT: The plan is a regular decagonal hemispherical triangulation. The elevation rises from a planar ten-node base to one apex at 72 inches.

## G-201 / PARTS BEFORE ASSEMBLY

## Every timber axis and node kit

Piece IDs match the interactive model. Every timber length below is node-center geometry only - not a blank, shoulder, tip-to-tip, or cut length.

| FAMILY    | QTY | NODE-CENTER AXIS     | INCIDENCE                                  |
|-----------|-----|----------------------|--------------------------------------------|
| Short S   | 30  | 39.350380 in         | 10 H4-H5 + 20 H5-H6                        |
| Long L    | 35  | 44.498447 in         | 10 H4-H4 + 10 H4-H6 + 15 H6-H6             |
| Node kits | 26  | 10 H4 / 6 H5 / 10 H6 | Topology families; not machining templates |

| SHORT PIECES   | LONG PIECES    | NODE KITS     |
|----------------|----------------|---------------|
| S-01 V001-V002 | L-01 V002-V003 | V001 H5 ABOVE |
| S-02 V001-V003 | L-02 V002-V006 | V002 H6 ABOVE |
| S-03 V001-V004 | L-03 V002-V007 | V003 H6 ABOVE |
| S-04 V001-V005 | L-04 V002-V011 | V004 H6 ABOVE |
| S-05 V001-V006 | L-05 V003-V004 | V005 H6 ABOVE |
| S-06 V002-V012 | L-06 V003-V007 | V006 H6 ABOVE |
| S-07 V003-V013 | L-07 V003-V008 | V007 H6 ABOVE |
| S-08 V004-V014 | L-08 V004-V005 | V008 H6 ABOVE |
| S-09 V005-V015 | L-09 V004-V008 | V009 H6 ABOVE |
| S-10 V006-V016 | L-10 V004-V009 | V010 H6 ABOVE |
| S-11 V007-V012 | L-11 V005-V006 | V011 H6 ABOVE |
| S-12 V007-V013 | L-12 V005-V009 | V012 H5 ABOVE |
| S-13 V008-V013 | L-13 V005-V010 | V013 H5 ABOVE |
| S-14 V008-V014 | L-14 V006-V010 | V014 H5 ABOVE |
| S-15 V009-V014 | L-15 V006-V011 | V015 H5 ABOVE |
| S-16 V009-V015 | L-16 V007-V017 | V016 H5 ABOVE |
| S-17 V010-V015 | L-17 V007-V018 | V017 H4 BASE  |
| S-18 V010-V016 | L-18 V008-V019 | V018 H4 BASE  |
| S-19 V011-V012 | L-19 V008-V020 | V019 H4 BASE  |
| S-20 V011-V016 | L-20 V009-V021 | V020 H4 BASE  |
| S-21 V012-V017 | L-21 V009-V022 | V021 H4 BASE  |
| S-22 V012-V026 | L-22 V010-V023 | V022 H4 BASE  |
| S-23 V013-V018 | L-23 V010-V024 | V023 H4 BASE  |
| S-24 V013-V019 | L-24 V011-V025 | V024 H4 BASE  |
| S-25 V014-V020 | L-25 V011-V026 | V025 H4 BASE  |
| S-26 V014-V021 | L-26 V017-V018 | V026 H4 BASE  |
| S-27 V015-V022 | L-27 V017-V026 |               |
| S-28 V015-V023 | L-28 V018-V019 |               |
| S-29 V016-V024 | L-29 V019-V020 |               |
| S-30 V016-V025 | L-30 V020-V021 |               |
|                | L-31 V021-V022 |               |
|                | L-32 V022-V023 |               |
|                | L-33 V023-V024 |               |
|                | L-34 V024-V025 |               |
|                | L-35 V025-V026 |               |

**MATERIAL BASIS:** Douglas fir, nominal 2 x 2 in, modeled dressed section 1.500 x 1.500 in. Confirm actual stock, moisture, grade, defects, and grain before any fabrication release.

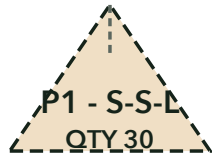
## P-101 / GROSS WOOD FACE SCHEDULE

## Two exact face families

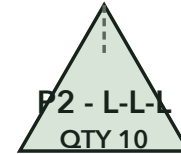
These triangles are derived from the canonical node-center faces. Gross total: 30238.094167 sq in / 209.986765 sq ft. They are not finished panel cuts or purchase quantities.

### P-101 GROSS NODE-CENTER FACE FAMILIES - NOT CUT TEMPLATES

SIDES 39.350380 / 39.350380 IN  
BASE 44.498447 · ALT 32.456501 IN



SIDES 44.498447 IN EACH  
ALT 38.536786 IN



**GROSS MODEL FACES ONLY - EDGE BUILD-UP, CUTBACKS, JOINTS, GASKETS, AND DRAINAGE ARE UNSET**

| TYPE     | QTY | GROSS SIDES                             | ANGLES                                   | AREA EACH           |
|----------|-----|-----------------------------------------|------------------------------------------|---------------------|
| P1 S-S-L | 30  | 39.350380 / 39.350380 /<br>44.498447 in | 55.569012 / 55.569012<br>/ 68.861976 DEG | 722.131952<br>sq in |
| P2 L-L-L | 10  | 44.498447 in equilateral                | 60 / 60 / 60 DEG                         | 857.413562<br>sq in |

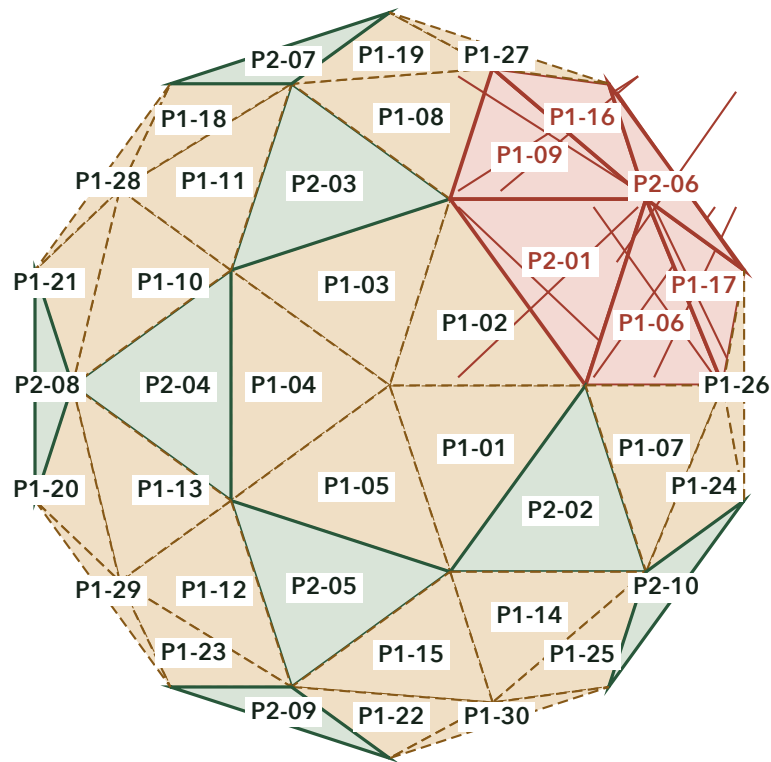
| P1 PIECE / FACE / VERTICES | P2 PIECE / FACE / VERTICES |
|----------------------------|----------------------------|
| P1-01 F001 V001-V002-V006  | P2-01 F006 V002-V003-V007  |
| P1-02 F002 V001-V003-V002  | P2-02 F008 V002-V011-V006  |
| P1-03 F003 V001-V004-V003  | P2-03 F010 V003-V004-V008  |
| P1-04 F004 V001-V005-V004  | P2-04 F013 V004-V005-V009  |
| P1-05 F005 V001-V006-V005  | P2-05 F016 V005-V006-V010  |
| P1-06 F007 V002-V007-V012  | P2-06 F023 V007-V018-V017  |
| P1-07 F009 V002-V012-V011  | P2-07 F026 V008-V020-V019  |
| P1-08 F011 V003-V008-V013  | P2-08 F029 V009-V022-V021  |
| P1-09 F012 V003-V013-V007  | P2-09 F032 V010-V024-V023  |
| P1-10 F014 V004-V009-V014  | P2-10 F035 V011-V026-V025  |
| P1-11 F015 V004-V014-V008  |                            |
| P1-12 F017 V005-V010-V015  |                            |
| P1-13 F018 V005-V015-V009  |                            |
| P1-14 F019 V006-V011-V016  |                            |
| P1-15 F020 V006-V016-V010  |                            |
| P1-16 F021 V007-V013-V018  |                            |
| P1-17 F022 V007-V017-V012  |                            |
| P1-18 F024 V008-V014-V020  |                            |
| P1-19 F025 V008-V019-V013  |                            |
| P1-20 F027 V009-V015-V022  |                            |
| P1-21 F028 V009-V021-V014  |                            |
| P1-22 F030 V010-V016-V024  |                            |
| P1-23 F031 V010-V023-V015  |                            |
| P1-24 F033 V011-V012-V026  |                            |
| P1-25 F034 V011-V025-V016  |                            |
| P1-26 F036 V012-V017-V026  |                            |
| P1-27 F037 V013-V019-V018  |                            |
| P1-28 F038 V014-V021-V020  |                            |
| P1-29 F039 V015-V023-V022  |                            |
| P1-30 F040 V016-V025-V024  |                            |

## P-102 / PANEL PLACEMENT

## All 40 faces in one map

The top projection is a non-overlapping map of the canonical triangulated disk. Each printed label is the panel piece ID listed on P-101.

**P-102 40-FACE PLAN PLACEMENT MAP - NOT TO SCALE**



P1 = dashed isosceles

P2 = solid equilateral

X-MARKED = entrance-view omitted

| ENTRANCE-VIEW AFFECTED FACES            | PIECE CONSEQUENCE                                      | STATUS                            |
|-----------------------------------------|--------------------------------------------------------|-----------------------------------|
| F006 / F007 / F012 / F021 / F022 / F023 | Six canonical face panels hidden only in entrance view | No replacement enclosure designed |

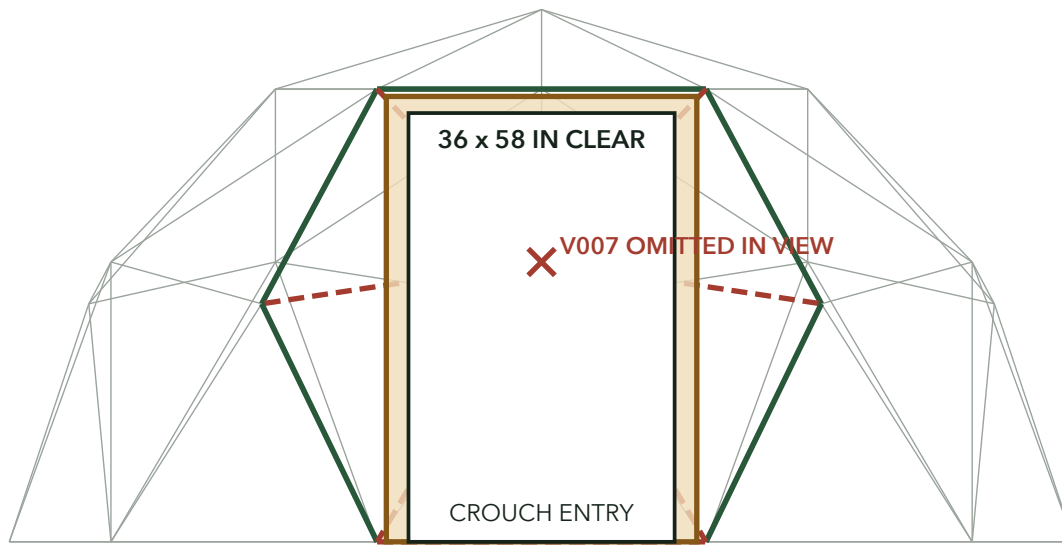
FIGURE TEXT EQUIVALENT: Thirty isosceles P1 faces and ten equilateral P2 faces tile the complete dome. Six marked faces form the optional entrance visualization patch.

## E-101 / OPTIONAL ENTRANCE STUDY

## Crouch opening aligned to one shell patch

The entrance is a visual opening study. It deliberately does not alter the canonical schedules or exports; affected shell parts are hidden only in this view.

### E-101 OPTIONAL ENTRANCE PATCH + CASSETTE - REPLACEMENT LOAD PATH NOT DESIGNED



DASHED = omitted canonical members

HEAVY SOLID = retained patch boundary

| ITEM              | VALUE                                          | BOUNDARY                                |
|-------------------|------------------------------------------------|-----------------------------------------|
| Clear opening     | 36 x 58 in                                     | Crouch entry; not full standing access  |
| Outer cassette    | 42.00 x 60.25 in                               | 3 in jambs / 2.25 in lintel study       |
| Source patch rise | 61.246858 in                                   | Geometric patch bound only              |
| Hidden members    | L-03 / L-06 / S-11 / S-12 / L-16 / L-17 / L-26 | Seven canonical members omitted in view |
| Retained boundary | L-01 / S-06 / S-07 / S-21 / S-23               | Five canonical boundary members remain  |
| Hidden node       | V007                                           | One canonical node omitted in view      |

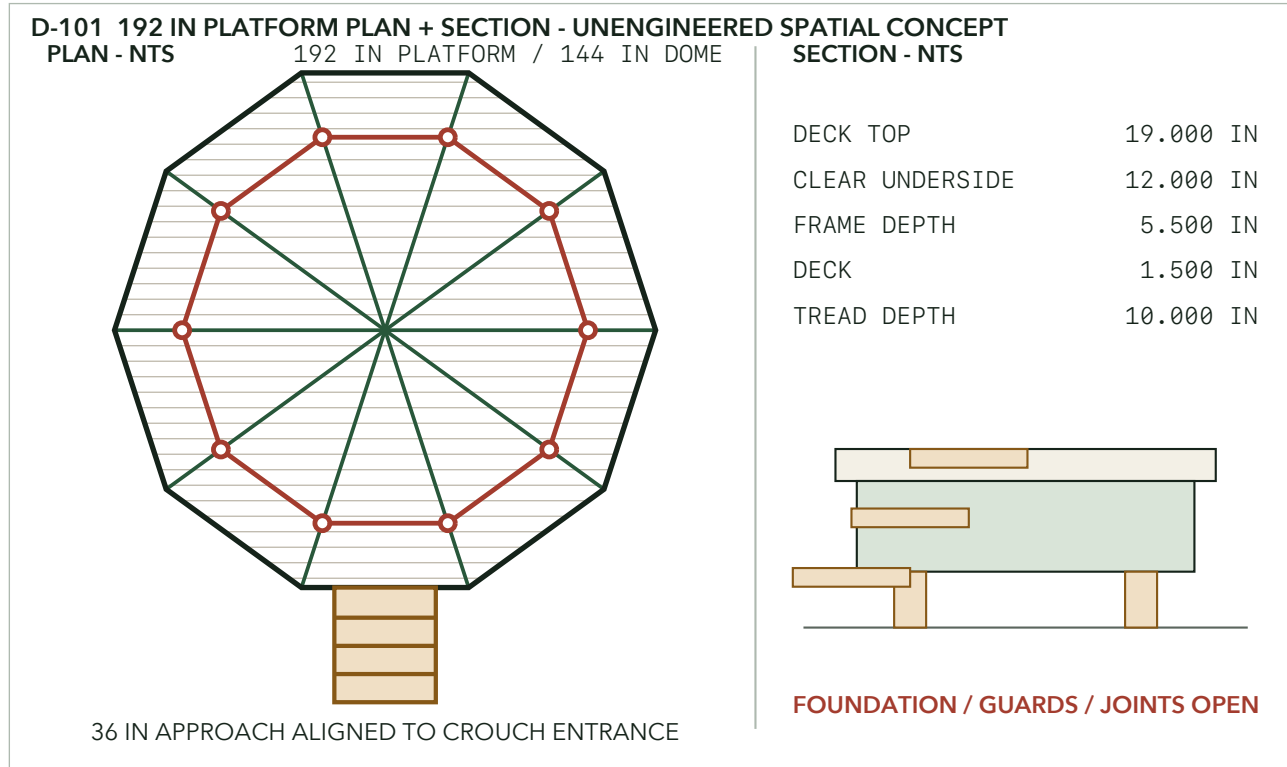
**VISUAL OPENING STUDY - REPLACEMENT LOAD PATH NOT DESIGNED. Reinforcement, joints, door, threshold, weather seals, drainage, egress, guards, foundation, and acoustic performance remain unevaluated.**

FIGURE TEXT EQUIVALENT: Dashed red members and node V007 are omitted from the entrance view. A 36 by 58 inch clear cassette is centered within the selected six-face patch.

## D-101 / PLATFORM + APPROACH

## Dome above a larger all-wood platform concept

The platform is spatially aligned with the 144 inch dome and 36 inch entrance. It is not a joist, footing, guard, or connection design.



| PARAMETER          | VALUE                                          | MODEL MEANING                     |
|--------------------|------------------------------------------------|-----------------------------------|
| Platform diameter  | 192.000 in                                     | Regular decagon, corner to corner |
| Dome base diameter | 144.000 in                                     | Node center, corner to corner     |
| Apron              | 24.000 in vertex / 22.825356 in flat           | Not 24 in everywhere              |
| Outer side         | 59.331263 in                                   | Ten equal rim segments            |
| Deck + frame       | 35 boards / 10 radial / 10 rim                 | Counts only; sizes unengineered   |
| Supports           | 10 + one center hub                            | Concept axes only                 |
| Vertical           | 19.000 in deck top / 12.000 in clear underside | Three-rise approach; 10 in treads |

**ALL-WOOD CONNECTION GOAL ONLY.** The H4 clearance body reaches 1.500 in below the base-node datum, so a recess or raised bearing datum is unresolved. Foundation, load path, isolation, weathering, guards, joints, and acoustics are not modeled.

FIGURE TEXT EQUIVALENT: A 192 inch decagonal deck surrounds the 144 inch dome, with ten radial frame lines, ten supports at dome base-node axes, and a centered three-step approach.

## J-301 / PORT-NORMAL CLEARANCE

## Tested digital envelopes clear; structural design open

Each shoulder face is normal to its member axis at one fixed setback. The two-shell body and pocket volumes are explicit digital envelopes, not selected or capacity-checked joinery.

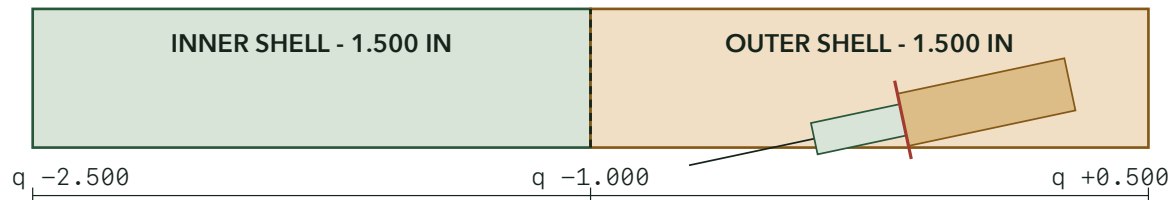
CURRENT JOINT STUDY

### TESTED ENVELOPES CLEAR - STRUCTURAL REVIEW OPEN

#### J-301 GENERIC SINGLE-PORT SECTION - SPATIAL CLEARANCE ONLY

SHOULDER FACE NORMAL TO MEMBER AXIS

TENON + OVERSIZED POCKET ENVELOPE



SCHEMATIC DATUM RELATIONSHIP - NOT A JOINT DETAIL

| INPUT               | VALUE                                        | AUDIT MEANING                           |
|---------------------|----------------------------------------------|-----------------------------------------|
| Shoulder            | Axis-normal at S = 4.000 in                  | Fixed port datum                        |
| Convex footprints   | H4 9.693632 / H5 10.110635 / H6 10.270510 in | Maximum planar diameters                |
| Tenon study         | 1.250 x 0.750 x 0.500 in                     | Integral volume only                    |
| Pocket envelope     | 1.300 x 0.780 x 0.530 in                     | Collision envelope; not fit tolerance   |
| Radial slab         | q = -2.500 to +0.500 in                      | Two 1.500 in shells split at q = -1.000 |
| Member roll         | width-radial · fixed                         | Fixed study condition                   |
| Sampled roll domain | 0 to 90 DEG / 1 DEG steps                    | 91 positions; robustness study          |

| CLEARANCE CHECK                | RESULT                      | MINIMUM               |
|--------------------------------|-----------------------------|-----------------------|
| Pocket pairs / 91 rolls        | 0 / 24,570 collide          | 1.396637 in           |
| 6 in member proxies / 91 rolls | 0 / 24,570 collide          | 1.537727 in           |
| Continuous pocket bound        | 0 / 270 collide             | 1.380331 in           |
| Continuous member bound        | 0 / 270 collide             | 1.408257 in           |
| Pocket split                   | Every pocket crosses q = -1 | 0.347980 in each side |
| Shoulder to other faces        | Every modeled shoulder fits | 1.011118 in           |

TEST DOMAIN: 24,570 = 270 installed port pairs x 91 one-degree rolls. The 6.000 in full-section proxies begin at each shoulder; they are not complete timbers.

TFEC SCOPE: These dimensions establish no TFEC 1-2019 standard tension-loaded wood-peg detail in the modeled 1.500 x 1.500 in section. Shear-loaded or alternative connections require separate engineering and test evidence.

**RELEASE BOUNDARY:** The superseded untrimmed 1.5 x 1.25 x 0.5 in tenon boxes intersected at every tested H4, H5, and H6 pair and remain rejected. No fit, retention, ligament, grain, adhesive, tolerance, moisture, strength, load, assembly, or durability approval is issued.

## R-001 / RELEASE GATE + AUDIT TRACE

## What Jantz can trust - and what remains open

Rev 08 consolidates the complete digital reference for Black Belt Building without converting planning geometry into construction authorization.

| VERIFIED / RECORDED                                                                                          | OPEN / WITHHELD                                                                                                         |
|--------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Class-I 2V topology, node coordinates, 65 axes, two chord classes, 26 node valences, and 40 face incidences. | Structural loads, member capacity, buckling, connection capacity, anchorage, foundation, code, permit, and occupancy.   |
| Gross P1/P2 triangle geometry and 209.986765 sq ft total mathematical face area.                             | Finished panel cuts, thickness, attachment, seals, drainage, fire, weathering, moisture movement, and acoustics.        |
| Entrance affected-set adjacency and 36 x 58 in visual opening envelope.                                      | Replacement shell load path, reinforcement, joints, door, threshold, egress, guards, and weather enclosure.             |
| 192 in platform spatial geometry, apron relationships, frame/support counts, and approach alignment.         | Joist/support sizing, bearing, footings, guards, all-wood joint dimensions, isolation, durability, and acoustic tuning. |

### Model assumptions

Geometry: All dimensions are generated from a 72 in spherical radius. Member lengths are node-center to node-center chords, not finished stock cuts. Rendered hub prisms are schematic topology markers; the full-length timber solids overlap them by construction. The 2 × 2 section is modeled at a dressed 1.5 × 1.5 in and must be checked against the actual stock.

Clearance model: The current study uses exact port-normal faces at a 4 in axial shoulder setback. The 24,570 sampled checks are 270 installed port pairs evaluated at 91 one-degree roll positions from 0 through 90 degrees. The sampled and continuous full-section member checks use 6 in external envelopes beginning at each shoulder, not complete timber lengths. The two-shell joint study clears the modeled member, tenon, pocket, key-relief, and center-bore envelopes, but is not structurally approved or build-ready.

Outside this model: The geometric cap is complete; loads, connections, foundations, openings, and code compliance are outside this model.

### Next gate

Confirm actual stock, species and grade, service moisture and exposure, site loads and occupancy, foundation and anchorage, hub material and layup, retention, assembly sequence, weather enclosure, acoustics, and tooling. Then complete exact solids, ligament and grain checks, engineered capacity calculations, a full-scale prototype/test program, and an explicit fabrication release.

### References + trace

Canonical sources: lib/geodesic.ts, lib/joinery.ts, and lib/spec.ts. Serialized by scripts/export\_pdf\_data.mjs; rendered by scripts/generate\_audit\_pdf.py. Model-data SHA-256:

6a270d1a33f110e6a227a0b6a629cfc6170df11bcb4fc31f2726bd10dae6f403.

Live accessible model: monroe-domes.vercel.app. Timber reference: TFEC 1-2019, Section 3.4 tension-joint provisions. Project-specific engineering remains required.

| REVISION | DATE       | RECORD                                                            |
|----------|------------|-------------------------------------------------------------------|
| 06       | 2026-08-27 | Core centerline and joint-clearance audit                         |
| 07       | 2026-08-29 | Field reference: parts, panels, entrance, and platform            |
| 08       | 2026-09-09 | Explicit test domain, safer CSV, stronger audit, and 3V reference |

**NO APPROVAL OR SIGNATURE BLOCK. REFERENCE STUDY ONLY.**